



भारतीय विज्ञान शिक्षा एवं अनुसंधान संस्थान मोहाली

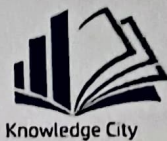
(मानव संसाधन विकास मंत्रालय का एक स्वायत्त संस्थान, भारत सरकार के अधीन)

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END-SEMESTER EXAM

EES301 Global biogeochemical cycles and Climate change

Date: 14.11.2025

Total Marks: 50

NAME:

ROLL NUMBER:

1. Why is the chemical weathering process more rapid and consistent in tropical forests than in deserts? (2 marks)
2. How are the congruent and incongruent reactions different in the carbonation-based weathering process? Explain with the help of specific examples (3 Marks)
3. Describe how the ocean pH influences the speciation of dissolved CO₂ in seawater? Explain the Ocean Acidification effects in the context of climate change. (3 marks)
4. Explain any three microbial N cycling processes that occur in soils with specific reactions. (3 marks)
5. Illustrate and explain the key processes in the redox cycle of Sulfur. (4 Marks)
6. Explain the source and fate of Dimethyl Sulfide in the atmosphere, along with its potential effect on global climate. (2 marks)
7. Describe how the pH influences the speciation of dissolved inorganic forms of phosphorus in seawater? (2 marks)
8. What is the primary ore/mineral and the most toxic form of mercury on Earth? How does the Hg – Methylation process occur in lake systems? (1+2 marks)
9. Despite its abundant presence in the Earth's crust, why is iron not readily available for organisms? What are the roles of microbes in the biogeochemical cycling of iron? Explain with the key processes. (1+3 marks)
10. Explain the terms Forcing, Response, and Feedback with any example in the context of the climate system. (3 marks)
11. How do the biological processes in biogeochemical cycles affect the global radiative balance? Explain with any example from the terrestrial/marine system. (2 marks)
12. How are the biogeochemical cycles of carbon and nitrogen coupled? Explain the interdependency with specific examples. (3 marks)
13. How does the large-scale eutrophication lead to the creation of dead zones in aquatic systems (2 marks)
14. The annual CO₂ influx of an ocean is observed to be 20 Gt CO₂. Only 20% of the absorbed influx is permanently sequestered or forms a carbon deposit in the deep ocean water, and the remaining 80% is eventually recycled back. Under this scenario, how many grams of carbon (total C) will be deposited in the ocean over a period of 10 years? (3 Marks)

15. The soil reservoir has 100 Tg P in 2020. The average annual flux rates of Weathering (inflow), River transport or runoff to waterways (outflow), and Fertilizer (inflow) of the soil pool are 5, 7, and 3 Tg P, respectively. What will be the P content of the soil pool in 2025? (2 Marks)

MCQs

1. Which one of the following is not a characteristic difference between the processes of Weathering and Erosion?
 - a. Weathering is described as a static process, while Erosion is a dynamic process
 - b. Weathering involves only physical/mechanical processes, while Erosion involves water/wind/snow processes
 - c. Weathering is the breaking down/decomposition of rocks, while Erosion is the displacement of solids
 - d. Weathering plays an important role in soil formation, while Erosion is considered a destructive process for soil
2. Which soil horizon/zone/layer has the highest microbial activity?
 - a. B
 - b. A
 - c. O
 - d. C
3. In the redox cycle for carbon, which group of processes represents 'autotrophy', i.e., the conversion of inorganic carbon (CO_2) into organic compounds?
 - a. Photosynthesis and Chemolithoautotrophy
 - b. Methanogenesis and Methanotrophy
 - c. Acetogenesis and Fermentation
 - d. Photosynthesis and Methanogenesis
4. Which of the following is not an observed effect or consequence of acid rain resulting from the global sulphur cycle?
 - a. Water and soil acidification
 - b. Nutrient-deficient soils
 - c. Corrosion of infrastructure
 - d. Eutrophication
5. The microbial processes of Ferric iron reduction and Sulfate reduction are fundamentally similar because both:
 - a. Are forms of anaerobic respiration where the oxidized compound acts as the terminal electron acceptor
 - b. Occur exclusively under oxic (aerobic) conditions
 - c. Utilize the reactant as an electron donor
 - d. Directly produce highly insoluble minerals that precipitate and settle

6. The oxidation of Hydrogen Sulfide is carried out by different microbes under two distinct environmental conditions. Which pair correctly matches the condition with the primary product and organism type?
- a. Oxidic; Sulfate; Anoxygenic phototrophs
 - b. Anoxic; Elemental Sulfur; Anoxygenic phototrophs
 - c. Anoxic; Sulfate; Chemolithotrophic SOB
 - d. Oxidic; Elemental Sulfur; Sulfate-reducing bacteria
7. In an anoxic (anaerobic) sediment environment, what is the role of Manganese that is analogous to the role of Ferric Iron?
- a. Mn^{2+} is oxidized to Mn^{4+} by nitrifying bacteria in the absence of oxygen.
 - b. Mn^{4+} serves as an electron acceptor for microbes, reducing it to soluble Mn^{2+}
 - c. Mn minerals combine with H_2S to form insoluble sulfides
 - d. Mn^{3+} species is the most stable and soluble form
8. State the Nitrogen reservoirs on Earth in descending order according to the N concentration/content percentage.
- a. Geological > Atmosphere > Oceans > Biosphere
 - b. Atmosphere > Geological > Oceans > Biosphere
 - c. Oceans > Biosphere > Atmosphere > Geological
 - d. Biosphere > Atmosphere > Geological > Oceans
9. Select the odd one among the following as far as the sources and sinks of methane are concerned.
- a. Ruminant gut
 - b. Atmosphere
 - c. Paddy soils
 - d. Landfills